

Comprehensive Math Refresher & Reference Guide

Audience: Experienced engineer. BS+ background in math through differential equations and linear algebra.

Purpose: Dense reference for solving any standard problem in Algebra, Calculus 1–3, Differential Equations, and Linear Algebra.

Format: Every formula defined. Every technique includes a worked numerical example. Every section includes a Common Mistakes callout.

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SUBJECT 1: ALGEBRA

1.1 Factoring and Algebraic Manipulation

Difference of Squares

$$a^2 - b^2 = (a + b)(a - b)$$

Example: $x^2 - 9 = (x + 3)(x - 3)$

Perfect Square Trinomial

$$a^2 \pm 2ab + b^2 = (a \pm b)^2$$

Example: $x^2 + 6x + 9 = (x + 3)^2$

Sum/Difference of Cubes

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

Example: $8x^3 - 27 = (2x - 3)(4x^2 + 6x + 9)$

General Trinomial Factoring

For $ax^2 + bx + c$: find two numbers that multiply to ac and add to b . Split the middle term and factor by grouping.

Example: $6x^2 + 7x - 3$. Need two numbers multiplying to $6 \cdot (-3) = -18$, adding to 7: that's 9 and -2 .

$$6x^2 + 9x - 2x - 3 = 3x(2x + 3) - 1(2x + 3) = (3x - 1)(2x + 3)$$

Factoring by Grouping

Example: $x^3 + 2x^2 - 3x - 6 = x^2(x + 2) - 3(x + 2) = (x^2 - 3)(x + 2)$

Completing the Square

To convert $ax^2 + bx + c$:

1. Factor out a from the x terms: $a \left(x^2 + \frac{b}{a}x \right) + c$
2. Add and subtract $\left(\frac{b}{2a} \right)^2$ inside: $a \left(x + \frac{b}{2a} \right)^2 + c - \frac{b^2}{4a}$

Example: $2x^2 - 8x + 5$

$$= 2(x^2 - 4x) + 5 = 2(x^2 - 4x + 4 - 4) + 5 = 2(x - 2)^2 - 8 + 5 = 2(x - 2)^2 - 3$$

Use cases: vertex form of parabola, trig substitution setup, integrating $\int \frac{dx}{x^2 + bx + c}$.

Polynomial Long Division

Divide $2x^3 - 3x^2 + x - 5$ by $x - 2$:

$$\begin{array}{r}
 2x^2 + x + 3 \\
 x - 2 \overline{) 2x^3 - 3x^2 + x - 5} \\
 \underline{2x^3 - 4x^2} \\
 x^2 + x \\
 \underline{x^2 - 2x} \\
 3x - 5 \\
 \underline{3x - 6} \\
 1
 \end{array}$$

Result: $2x^2 + x + 3 + \frac{1}{x-2}$

Synthetic Division

Only for dividing by $(x - c)$. For $2x^3 - 3x^2 + x - 5 \div (x - 2)$, use $c = 2$:

$$\begin{array}{r|rrrr}
 2 & 2 & -3 & 1 & -5 \\
 & & 4 & 2 & 6 \\
 \hline
 & 2 & 1 & 3 & 1
 \end{array}$$

$$\begin{array}{r} | \text{-----} \\ 2 \quad 1 \quad 3 \quad 1 \end{array}$$

Same result: $2x^2 + x + 3$ remainder 1.

Partial Fraction Decomposition

Step 1: If $\deg(\text{numerator}) \geq \deg(\text{denominator})$, do polynomial long division first.

Step 2: Factor denominator completely over the reals.

Step 3: Set up the decomposition:

Factor in denominator	Contribution to decomposition
$(ax + b)$	$\frac{A}{ax + b}$
$(ax + b)^2$	$\frac{A}{ax + b} + \frac{B}{(ax + b)^2}$
$(ax + b)^n$	$\frac{A_1}{ax + b} + \frac{A_2}{(ax + b)^2} + \cdots + \frac{A_n}{(ax + b)^n}$
$ax^2 + bx + c$ (irreducible)	$\frac{Ax + B}{ax^2 + bx + c}$

Example (distinct linear): $\frac{3x + 1}{(x - 1)(x + 2)}$

$$\frac{3x + 1}{(x - 1)(x + 2)} = \frac{A}{x - 1} + \frac{B}{x + 2}$$

Multiply through: $3x + 1 = A(x + 2) + B(x - 1)$

Set $x = 1$: $4 = 3A \Rightarrow A = \frac{4}{3}$

Set $x = -2$: $-5 = -3B \Rightarrow B = \frac{5}{3}$

Example (irreducible quadratic): $\frac{x^2 + 2x + 3}{(x + 1)(x^2 + 4)}$

$$= \frac{A}{x + 1} + \frac{Bx + C}{x^2 + 4}$$

Multiply through: $x^2 + 2x + 3 = A(x^2 + 4) + (Bx + C)(x + 1)$

Set $x = -1$: $1 - 2 + 3 = A(5) \Rightarrow A = \frac{2}{5}$

Expand and match coefficients for B and C .

Common Mistakes:

- Forgetting to do long division when $\deg(\text{num}) \geq \deg(\text{denom})$
- Setting up $\frac{Ax+B}{(ax+b)^2}$ for a repeated linear factor instead of two separate terms
- Skipping the irreducible quadratic check — $x^2 + 4$ does NOT factor over the reals

1.2 Exponent and Logarithm Laws

Exponent Laws

Law	Formula
Product	$a^m \cdot a^n = a^{m+n}$
Quotient	$a^m / a^n = a^{m-n}$
Power	$(a^m)^n = a^{mn}$
Product base	$(ab)^n = a^n b^n$
Negative	$a^{-n} = 1/a^n$
Zero	$a^0 = 1 \ (a \neq 0)$
Fractional	$a^{m/n} = \sqrt[n]{a^m} = (\sqrt[n]{a})^m$

Logarithm Laws

$$\log_b(xy) = \log_b x + \log_b y$$

$$\log_b\left(\frac{x}{y}\right) = \log_b x - \log_b y$$

$$\log_b(x^r) = r \log_b x$$

$$\log_b x = \frac{\ln x}{\ln b} \quad (\text{change of base})$$

$$\ln(e^x) = x, \quad e^{\ln x} = x$$

$$\log_b(b^x) = x, \quad b^{\log_b x} = x$$

Common Mistakes:

- $\ln(x + y) \neq \ln x + \ln y$ (no product rule for sums)
- $(\ln x)^2 \neq 2 \ln x$ (the power rule requires the argument to be raised, not the log itself)

1.3 Solving Equations

Quadratic: All Methods

Given $ax^2 + bx + c = 0$:

Quadratic Formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Discriminant $\Delta = b^2 - 4ac$: if $\Delta > 0$, two real roots; $\Delta = 0$, one repeated real root; $\Delta < 0$, two complex roots.

Example: $3x^2 - 5x - 2 = 0$

$$x = \frac{5 \pm \sqrt{25 + 24}}{6} = \frac{5 \pm 7}{6}$$

$$x = 2 \text{ or } x = -\frac{1}{3}$$

Rational Equations

Multiply through by LCD. Check for extraneous solutions (values making the original denominator zero).

Example: $\frac{1}{x-2} + \frac{3}{x+1} = \frac{4}{x^2-x-2}$

Note $x^2 - x - 2 = (x - 2)(x + 1)$. Multiply by $(x - 2)(x + 1)$:

$$(x + 1) + 3(x - 2) = 4$$

$$x + 1 + 3x - 6 = 4 \Rightarrow 4x = 9 \Rightarrow x = 9/4$$

Check: $x = 9/4$ doesn't make any denominator zero. ✓

Radical Equations

Isolate the radical, square both sides, check for extraneous solutions.

Example: $\sqrt{2x + 3} = x - 1$

Square: $2x + 3 = x^2 - 2x + 1 \Rightarrow x^2 - 4x - 2 = 0$

$$x = \frac{4 \pm \sqrt{16 + 8}}{2} = 2 \pm \sqrt{6}$$

Check both in original: $x = 2 + \sqrt{6}$ works; $x = 2 - \sqrt{6} \approx -0.449$ gives $\sqrt{} \approx 1.55$ but $x - 1 \approx -1.449$ — extraneous.

Absolute Value Equations

$|f(x)| = c$: if $c > 0$, solve $f(x) = c$ and $f(x) = -c$.

$|f(x)| = |g(x)|$: solve $f(x) = g(x)$ and $f(x) = -g(x)$.

Exponential Equations

Isolate the exponential, apply $\ln$ to both sides.

Example: $3^{2x-1} = 5$

$$(2x - 1) \ln 3 = \ln 5$$

$$x = \frac{\ln 5 / \ln 3 + 1}{2} = \frac{\log_3 5 + 1}{2}$$

Logarithmic Equations

Isolate the log, exponentiate. Check domain (argument must be positive).

Example: $\ln(x + 2) + \ln(x - 1) = \ln(4x)$

$$\ln[(x + 2)(x - 1)] = \ln(4x)$$

$$(x + 2)(x - 1) = 4x$$

$$x^2 + x - 2 = 4x \Rightarrow x^2 - 3x - 2 = 0$$

$$x = \frac{3 \pm \sqrt{17}}{2}$$

Discard negative root (check domain).

1.4 Inequalities

Linear Inequalities

Solve like equations; flip the inequality when multiplying/dividing by a negative number.

Quadratic Inequalities

Solve $ax^2 + bx + c > 0$: find roots, test intervals.

Example: $x^2 - 5x + 6 > 0 \Rightarrow (x - 2)(x - 3) > 0$

Roots: $x = 2, 3$. Test intervals $(-\infty, 2)$, $(2, 3)$, $(3, \infty)$:

- $x = 0$: $(-2)(-3) = 6 > 0 \checkmark$
 - $x = 2.5$: $(0.5)(-0.5) < 0 \times$
 - $x = 4$: $(2)(1) = 2 > 0 \checkmark$
- Solution: $(-\infty, 2) \cup (3, \infty)$

Rational Inequalities

Find critical values (zeros of numerator and denominator), test intervals, exclude denominator zeros from solution.

Absolute Value Inequalities

- $|f(x)| < c$: $-c < f(x) < c$
- $|f(x)| > c$: $f(x) < -c$ or $f(x) > c$

Common Mistakes:

- Not flipping the inequality when multiplying by a negative
- Including denominator zeros in the solution set of a rational inequality

1.5 Functions

Domain Determination

- Rational: exclude where denominator = 0
- Square roots (even): radicand ≥ 0
- Logarithms: argument > 0
- Compositions: apply restrictions sequentially

Example: Domain of $f(x) = \sqrt{\ln(x - 1)}$

Need $x - 1 > 0$ (for $\ln$), so $x > 1$. Then need $\ln(x - 1) \geq 0$, so $x - 1 \geq 1$, so $x \geq 2$.

Domain: $[2, \infty)$

Inverse Functions

1. Verify f is one-to-one (horizontal line test, or confirm strictly monotone)
2. Write $y = f(x)$, swap x and y , solve for y
3. The result is $f^{-1}(x)$

Example: $f(x) = \frac{2x-3}{x+1}$. Swap: $x = \frac{2y-3}{y+1}$

$$x(y+1) = 2y-3 \Rightarrow xy+x = 2y-3 \Rightarrow y(x-2) = -3-x$$

$f^{-1}(x) = \frac{-x-3}{x-2}$

Transformations

Starting from $y = f(x)$:

Transformation	Effect
$y = f(x) + k$	Shift up k
$y = f(x - h)$	Shift right h
$y = -f(x)$	Reflect over x -axis
$y = f(-x)$	Reflect over y -axis
$y = af(x), a > 1$	Vertical stretch by a
$y = f(bx), b > 1$	Horizontal compress by b

1.6 Trigonometry — Complete Reference

Unit Circle (Key Values)

θ	$\sin \theta$	$\cos \theta$	$\tan \theta$
0	0	1	0
$\pi/6$	1/2	$\sqrt{3}/2$	$1/\sqrt{3}$
$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$	1
$\pi/3$	$\sqrt{3}/2$	1/2	$\sqrt{3}$
$\pi/2$	1	0	undefined
π	0	-1	0
$3\pi/2$	-1	0	undefined
2π	0	1	0

Reciprocal functions: $\csc = 1/\sin$, $\sec = 1/\cos$, $\cot = 1/\tan$

Pythagorean Identities

$\sin^2 \theta + \cos^2 \theta = 1$

$1 + \tan^2 \theta = \sec^2 \theta$

$1 + \cot^2 \theta = \csc^2 \theta$

Angle Addition / Subtraction

$\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta$

$$\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta$$

$$\tan(\alpha \pm \beta) = \frac{\tan \alpha \pm \tan \beta}{1 \mp \tan \alpha \tan \beta}$$

Double Angle Formulas

$$\sin(2\theta) = 2 \sin \theta \cos \theta$$

$$\cos(2\theta) = \cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta - 1 = 1 - 2 \sin^2 \theta$$

$$\tan(2\theta) = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

Power Reduction (used constantly in integration)

$$\sin^2 \theta = \frac{1 - \cos(2\theta)}{2}, \quad \cos^2 \theta = \frac{1 + \cos(2\theta)}{2}$$

Half Angle Formulas

$$\sin \frac{\theta}{2} = \pm \sqrt{\frac{1 - \cos \theta}{2}}, \quad \cos \frac{\theta}{2} = \pm \sqrt{\frac{1 + \cos \theta}{2}}$$

Sign depends on the quadrant of $\theta/2$.

Product-to-Sum

$$\sin A \cos B = \frac{1}{2} [\sin(A + B) + \sin(A - B)]$$

$$\cos A \cos B = \frac{1}{2} [\cos(A - B) + \cos(A + B)]$$

$$\sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)]$$

Inverse Trig Functions

Function	Domain	Range
$\arcsin x$	$[-1, 1]$	$[-\pi/2, \pi/2]$
$\arccos x$	$[-1, 1]$	$[0, \pi]$
$\arctan x$	$(-\infty, \infty)$	$(-\pi/2, \pi/2)$
$\operatorname{arcsec} x$	$\mathbb{R} \setminus (-1, 1)$	$\mathbb{R} \setminus (-\pi/2, \pi/2)$

Composition identities:

$$\sin(\arctan x) = \frac{x}{\sqrt{1+x^2}}, \quad \cos(\arctan x) = \frac{1}{\sqrt{1+x^2}}$$

$$\tan(\arcsin x) = \frac{x}{\sqrt{1-x^2}}$$

Draw a reference triangle to derive these.

Solving Trig Equations

Find all solutions in $[0, 2\pi)$, then add $2\pi k$ for general solution.

Example: $2 \sin^2 x - \sin x - 1 = 0$

Factor: $(2 \sin x + 1)(\sin x - 1) = 0$

$\sin x = -1/2 \Rightarrow x = 7\pi/6, 11\pi/6$

$\sin x = 1 \Rightarrow x = \pi/2$

Common Mistakes:

- Dividing by $\sin x$ or $\cos x$ (you lose solutions where they equal zero — factor instead)
- Forgetting that $\arcsin$ only returns values in $[-\pi/2, \pi/2]$; there's a second solution in $[\pi/2, 3\pi/2]$

1.7 Complex Numbers

Rectangular Form: $z = a + bi$, where $i = \sqrt{-1}$, $i^2 = -1$

- **Addition:** $(a + bi) + (c + di) = (a + c) + (b + d)i$
- **Multiplication:** $(a + bi)(c + di) = (ac - bd) + (ad + bc)i$
- **Division:** $\frac{a+bi}{c+di} = \frac{(a+bi)(c-di)}{c^2+d^2}$
- **Conjugate:** $\bar{z} = a - bi$; note $z\bar{z} = a^2 + b^2 = |z|^2$

Polar Form

$$z = r(\cos \theta + i \sin \theta) = re^{i\theta}$$

where $r = |z| = \sqrt{a^2 + b^2}$ (modulus) and $\theta = \arg(z) = \arctan(b/a)$ (argument, careful with quadrant).

Euler's formula: $e^{i\theta} = \cos \theta + i \sin \theta$

De Moivre's Theorem

$$(re^{i\theta})^n = r^n e^{in\theta} = r^n (\cos(n\theta) + i \sin(n\theta))$$

Example: $(1 + i)^8$. Write $1 + i = \sqrt{2}e^{i\pi/4}$.

$$(1 + i)^8 = (\sqrt{2})^8 e^{i \cdot 8\pi/4} = 16e^{2\pi i} = 16$$

n th Roots

The n n th roots of $re^{i\theta}$ are:

$$z_k = r^{1/n} e^{i(\theta+2\pi k)/n}, \quad k = 0, 1, \dots, n-1$$

SUBJECT 2: CALCULUS 1

2.1 Limits

Limit Laws

If $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$, then:

- $\lim[f \pm g] = L \pm M$
- $\lim[f \cdot g] = L \cdot M$
- $\lim[f/g] = L/M$ (provided $M \neq 0$)
- $\lim[f^n] = L^n$
- $\lim[\sqrt[n]{f}] = \sqrt[n]{L}$ (provided $L > 0$ for even n)

Formal Definition (ϵ - δ)

$\lim_{x \rightarrow a} f(x) = L$ means: for every $\epsilon > 0$, there exists $\delta > 0$ such that $0 < |x - a| < \delta \Rightarrow |f(x) - L| < \epsilon$.

Proof example: $\lim_{x \rightarrow 3} (2x - 1) = 5$.

$|f(x) - 5| = |2x - 1 - 5| = 2|x - 3|$. Set $2|x - 3| < \epsilon$, so $\delta = \epsilon/2$.

Techniques for Evaluating Limits

Direct substitution: try $f(a)$ first. If defined and continuous, you're done.

Factoring/canceling (0/0):

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} = \lim_{x \rightarrow 2} \frac{(x - 2)(x + 2)}{x - 2} = \lim_{x \rightarrow 2} (x + 2) = 4$$

Rationalization:

$$\lim_{x \rightarrow 0} \frac{\sqrt{x+4} - 2}{x} \cdot \frac{\sqrt{x+4} + 2}{\sqrt{x+4} + 2} = \lim_{x \rightarrow 0} \frac{x}{x(\sqrt{x+4} + 2)} = \frac{1}{4}$$

Squeeze Theorem: If $g(x) \leq f(x) \leq h(x)$ near a and $\lim g = \lim h = L$, then $\lim f = L$.

Example: $\lim_{x \rightarrow 0} x^2 \sin(1/x)$. Since $-1 \leq \sin(1/x) \leq 1$: $-x^2 \leq x^2 \sin(1/x) \leq x^2$. Both squeeze limits = 0, so the limit = 0.

Limits at Infinity

- $\lim_{x \rightarrow \infty} x^n = \infty$ for $n > 0$; $\lim_{x \rightarrow \infty} x^{-n} = 0$
- For rational functions: divide numerator and denominator by the highest power of x in the denominator.

Example: $\lim_{x \rightarrow \infty} \frac{3x^2 - 2x}{5x^2 + 1} = \lim_{x \rightarrow \infty} \frac{3 - 2/x}{5 + 1/x^2} = \frac{3}{5}$

Indeterminate Forms and Resolutions

Form	Resolution
$0/0$	Factor, rationalize, or L'Hôpital
∞/∞	Divide by dominant term or L'Hôpital
$0 \cdot \infty$	Rewrite as $0/(1/\infty)$ or $\infty/(1/0)$, then L'Hôpital
$\infty - \infty$	Combine fractions, rationalize
$0^0, 1^\infty, \infty^0$	Take $\ln$, convert to $0/0$ or ∞/∞ , then L'Hôpital

L'Hôpital's Rule: If $\lim f/g$ is $0/0$ or $\pm\infty/\infty$, then $\lim \frac{f(x)}{g(x)} = \lim \frac{f'(x)}{g'(x)}$ (provided the latter exists).

Example: $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = \lim_{x \rightarrow 0} \frac{e^x}{1} = 1$

Example (1^∞): $\lim_{x \rightarrow 0^+} x^x$. Let $L = \lim x^x$. Then $\ln L = \lim x \ln x = \lim \frac{\ln x}{1/x} \stackrel{H}{=} \lim \frac{1/x}{-1/x^2} = \lim(-x) = 0$. So $L = e^0 = 1$.

Continuity

f is continuous at a iff:

1. $f(a)$ is defined
2. $\lim_{x \rightarrow a} f(x)$ exists
3. $\lim_{x \rightarrow a} f(x) = f(a)$

Types of discontinuity:

- **Removable:** limit exists but $\neq f(a)$ (or $f(a)$ undefined)
- **Jump:** one-sided limits exist but are unequal
- **Infinite:** limit is $\pm\infty$

Intermediate Value Theorem: If f is continuous on $[a, b]$ and $f(a) < k < f(b)$, then $\exists c \in (a, b)$ with $f(c) = k$.

Common Mistakes:

- Applying L'Hôpital when the form is NOT $0/0$ or ∞/∞
- Forgetting to check that the post-L'Hôpital limit actually exists before concluding
- Assuming continuity at a point just because the function is "nice-looking" there

2.2 Derivatives — All Rules and Techniques

Limit Definition

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Example: Find $f'(x)$ for $f(x) = x^2 + 3x$.

$$\frac{(x+h)^2 + 3(x+h) - x^2 - 3x}{h} = \frac{2xh + h^2 + 3h}{h} = 2x + h + 3 \rightarrow 2x + 3$$

Differentiation Rules

Rule	Formula	Example
Power	$\frac{d}{dx}[x^n] = nx^{n-1}$	$\frac{d}{dx}[x^5] = 5x^4$
Constant multiple	$\frac{d}{dx}[cf] = cf'$	$\frac{d}{dx}[3x^4] = 12x^3$
Sum/Difference	$(f \pm g)' = f' \pm g'$	
Product	$(fg)' = f'g + fg'$	$(x^2 \sin x)' = 2x \sin x + x^2 \cos x$
Quotient	$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$	
Chain	$\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$	$\frac{d}{dx}[\sin(x^2)] = \cos(x^2) \cdot 2x$

Triple chain rule example: $\frac{d}{dx}[e^{\sin(x^2)}] = e^{\sin(x^2)} \cdot \cos(x^2) \cdot 2x$

Derivatives of Standard Functions

Function	Derivative
x^n	nx^{n-1}
e^x	e^x
a^x	$a^x \ln a$
$\ln x$	$1/x$
$\log_a x$	$1/(x \ln a)$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
$\cot x$	$-\csc^2 x$
$\sec x$	$\sec x \tan x$
$\csc x$	$-\csc x \cot x$
$\arcsin x$	$1/\sqrt{1-x^2}$
$\arccos x$	$-1/\sqrt{1-x^2}$
$\arctan x$	$1/(1+x^2)$
$\operatorname{arcsec} x$	$1/(\sqrt{x^2-1})$

Implicit Differentiation

Differentiate both sides with respect to x ; whenever you differentiate a y term, multiply by dy/dx .

Example: Find dy/dx for $x^3 + y^3 = 6xy$.

$$\begin{aligned} 3x^2 + 3y^2 \frac{dy}{dx} &= 6y + 6x \frac{dy}{dx} \\ \frac{dy}{dx}(3y^2 - 6x) &= 6y - 3x^2 \\ \frac{dy}{dx} &= \frac{6y - 3x^2}{3y^2 - 6x} = \frac{2y - x^2}{y^2 - 2x} \end{aligned}$$

Logarithmic Differentiation

Useful when: the function is a product/quotient of many terms, or has variable exponents.

Example: $y = \frac{x^3 \sqrt{x+1}}{(2x-1)^4}$

Take $\ln$: $\ln y = 3 \ln x + \frac{1}{2} \ln(x+1) - 4 \ln(2x-1)$

Differentiate: $\frac{y'}{y} = \frac{3}{x} + \frac{1}{2(x+1)} - \frac{8}{2x-1}$

Multiply both sides by y .

Example (variable exponent): $y = x^x$

$\ln y = x \ln x \Rightarrow \frac{y'}{y} = \ln x + 1 \Rightarrow y' = x^x (\ln x + 1)$

Higher-Order Derivatives

$f'', f''', f^{(n)}$ or $\frac{d^2 y}{dx^2}$, etc. — differentiate repeatedly.

Linear Approximation

$$L(x) = f(a) + f'(a)(x - a)$$

Example: Approximate $\sqrt{4.02}$. Let $f(x) = \sqrt{x}$, $a = 4$:

$f'(x) = 1/(2\sqrt{x})$, $f'(4) = 1/4$

$L(4.02) = 2 + \frac{1}{4}(0.02) = 2.005$

Common Mistakes:

- Product rule: writing $(fg)' = f'g'$ instead of $f'g + fg'$
- Chain rule: forgetting to multiply by the derivative of the inner function
- Implicit differentiation: not applying chain rule to y terms (missing the dy/dx factor)

2.3 Applications of Derivatives

Increasing/Decreasing and First Derivative Test

- $f' > 0$ on $(a, b) \rightarrow f$ is increasing on (a, b)
- $f' < 0$ on $(a, b) \rightarrow f$ is decreasing on (a, b)
- At a critical point c where $f'(c) = 0$ or $f'(c)$ DNE:
 - f' changes $+$ $\rightarrow$ $-$: local maximum
 - f' changes $-$ $\rightarrow$ $+$: local minimum
 - No sign change: neither

Concavity and Second Derivative Test

- $f'' > 0$: concave up
- $f'' < 0$: concave down
- $f''(c) = 0$ and sign changes: inflection point
- At a critical point c : $f''(c) > 0 \rightarrow$ local min; $f''(c) < 0 \rightarrow$ local max; $f''(c) = 0 \rightarrow$ inconclusive (use FDT)

Absolute Extrema on $[a, b]$ — Closed Interval Method

1. Find all critical points in (a, b)
2. Evaluate f at each critical point AND at a and b
3. Largest value = absolute max; smallest = absolute min

Example: $f(x) = x^3 - 3x^2 + 1$ on $[-1/2, 4]$

$$f'(x) = 3x^2 - 6x = 3x(x - 2); \text{ critical points } x = 0, 2$$

$$f(-1/2) = -1/8 - 3/4 + 1 = 1/8$$

$$f(0) = 1, f(2) = 8 - 12 + 1 = -3, f(4) = 64 - 48 + 1 = 17$$

Absolute max: 17 at $x = 4$. Absolute min: -3 at $x = 2$.

Applied Optimization — Setup Procedure

1. Assign variables, draw a diagram
2. Write the objective function (what you're maximizing/minimizing)
3. Write the constraint equation; use it to eliminate one variable
4. Differentiate, set $= 0$, solve
5. Verify it's a max/min (FDT, SDT, or endpoints)
6. Answer the original question (include units)

Example: A box with square base and open top, volume = 32 m^3 . Minimize surface area.

Let base side = x , height = h .

Surface area: $A = x^2 + 4xh$. Constraint: $x^2h = 32 \Rightarrow h = 32/x^2$.

$$A(x) = x^2 + 4x \cdot \frac{32}{x^2} = x^2 + \frac{128}{x}$$

$$A'(x) = 2x - \frac{128}{x^2} = 0 \Rightarrow x^3 = 64 \Rightarrow x = 4$$

$$h = 32/16 = 2. \text{ Dimensions: } 4 \times 4 \times 2 \text{ m.}$$

Related Rates — Setup Procedure

1. Draw and label all quantities (with values and rates given)
2. Write an equation relating the quantities
3. Differentiate both sides with respect to t
4. Substitute known values and solve for the unknown rate

Example: Ladder: 10 m ladder, foot sliding away at 2 m/s. How fast is the top sliding down when foot is 6 m from wall?

$$x^2 + y^2 = 100. \text{ Differentiate: } 2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0.$$

$$\text{At } x = 6: y = 8. \text{ Given } dx/dt = 2.$$

$$2(6)(2) + 2(8) \frac{dy}{dt} = 0 \Rightarrow \frac{dy}{dt} = -\frac{3}{2} \text{ m/s (negative = sliding down).}$$

Mean Value Theorem

If f is continuous on $[a, b]$ and differentiable on (a, b) , then $\exists c \in (a, b)$ such that:

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Rolle's Theorem: Special case where $f(a) = f(b)$, guaranteeing $f'(c) = 0$.

Common Mistakes:

- Optimization: not checking whether the critical point is a max or min (just assuming)
- Related rates: substituting numerical values BEFORE differentiating (rates must come from the differentiated equation)
- Closed interval method: forgetting to evaluate at endpoints

SUBJECT 3: CALCULUS 2

3.1 Integration — Foundations

Antiderivatives and Indefinite Integrals

$$\int f(x) dx = F(x) + C \text{ where } F'(x) = f(x)$$

Fundamental Theorem of Calculus

Part 1: $\frac{d}{dx} \int_a^x f(t) dt = f(x)$

With chain rule: $\frac{d}{dx} \int_a^{g(x)} f(t) dt = f(g(x)) \cdot g'(x)$

Part 2: $\int_a^b f(x) dx = F(b) - F(a)$

Standard Integral Table

$f(x)$	$\int f(x) dx$
$x^n \ (n \neq -1)$	$\frac{x^{n+1}}{n+1} + C$
$1/x$	$\ln x + C$
e^x	$e^x + C$
a^x	$\frac{a^x}{\ln a} + C$
$\sin x$	$-\cos x + C$
$\cos x$	$\sin x + C$
$\tan x$	$\ln \sec x + C$
$\cot x$	$\ln \sin x + C$
$\sec x$	$\ln \sec x + \tan x + C$

$f(x)$	$\int f(x) dx$
$\csc x$	$-\ln \csc x + \cot x + C$
$\sec^2 x$	$\tan x + C$
$\csc^2 x$	$-\cot x + C$
$\sec x \tan x$	$\sec x + C$
$\csc x \cot x$	$-\csc x + C$
$1/\sqrt{1-x^2}$	$\arcsin x + C$
$1/(1+x^2)$	$\arctan x + C$
$1/(x\sqrt{x^2-1})$	$\operatorname{arcsec} x + C$

3.2 Integration Techniques

u-Substitution

When to use: Composite function where $u = g(x)$ simplifies to a known form. Spot the pattern $f(g(x)) \cdot g'(x)$.

Procedure:

1. Choose $u = g(x)$
2. Compute $du = g'(x) dx$, solve for dx
3. Substitute everything — the result should be entirely in terms of u
4. Integrate in u
5. Back-substitute to x

Definite integrals: change limits: new limits are $g(a)$ and $g(b)$.

Example: $\int_0^1 2xe^{x^2} dx$

$u = x^2$, $du = 2x dx$; limits: $u(0) = 0$, $u(1) = 1$

$$\int_0^1 e^u du = e^1 - e^0 = e - 1$$

Example (tricky): $\int \frac{x}{x^2+1} dx$

$u = x^2 + 1$, $du = 2x dx$

$$\int \frac{1}{2u} du = \frac{1}{2} \ln |u| + C = \frac{1}{2} \ln(x^2 + 1) + C$$

Common Mistakes:

- Forgetting to change the limits for definite integrals
- Not fully substituting: leaving an x in the integrand
- Choosing u for the outer function instead of the inner

Integration by Parts

$$\int u dv = uv - \int v du$$

LIATE priority for u : Logarithm, Inverse trig, Algebraic, Trigonometric, Exponential

Example: $\int x e^x dx$

$$u = x, dv = e^x dx \rightarrow du = dx, v = e^x$$

$$= x e^x - \int e^x dx = x e^x - e^x + C = e^x(x - 1) + C$$

Example (repeated IBP): $\int x^2 \sin x dx$

Apply twice; tabular method:

Sign	u (differentiate)	dv (integrate)
+	x^2	$\sin x$
−	$2x$	$-\cos x$
+	2	$-\sin x$
−	0	$\cos x$

$$\text{Result: } -x^2 \cos x + 2x \sin x + 2 \cos x + C$$

Cycling IBP: $\int e^x \sin x dx$

$$u = e^x, dv = \sin x: = -e^x \cos x + \int e^x \cos x dx$$

$$\text{Apply again: } = -e^x \cos x + e^x \sin x - \int e^x \sin x dx$$

$$\text{Let } I = \int e^x \sin x dx: I = -e^x \cos x + e^x \sin x - I$$

$$2I = e^x(\sin x - \cos x) \Rightarrow I = \frac{e^x(\sin x - \cos x)}{2} + C$$

$$\int \ln x dx: u = \ln x, dv = dx: = x \ln x - x + C$$

Trigonometric Integrals

$$\int \sin^m x \cos^n x dx:$$

- If m (power of $\sin$) is odd: save one $\sin x$, convert rest using $\sin^2 x = 1 - \cos^2 x$, let $u = \cos x$
- If n (power of $\cos$) is odd: save one $\cos x$, convert rest using $\cos^2 x = 1 - \sin^2 x$, let $u = \sin x$
- If both even: use power reduction identities $\sin^2 = (1 - \cos 2x)/2$, $\cos^2 = (1 + \cos 2x)/2$

Example: $\int \sin^3 x \cos^2 x dx$

$$= \int \sin^2 x \cos^2 x \sin x dx = \int (1 - \cos^2 x) \cos^2 x \sin x dx$$

$$u = \cos x, du = -\sin x dx:$$

$$= -\int (1 - u^2) u^2 du = -\int (u^2 - u^4) du = -\frac{u^3}{3} + \frac{u^5}{5} + C = -\frac{\cos^3 x}{3} + \frac{\cos^5 x}{5} + C$$

$$\int \tan^m x \sec^n x dx:$$

- n even: save $\sec^2 x$, convert rest using $\sec^2 = 1 + \tan^2$, let $u = \tan x$
- m odd: save $\sec x \tan x$, convert rest using $\tan^2 = \sec^2 - 1$, let $u = \sec x$

Trigonometric Substitution

Integrand form	Substitution	Identity used
$\sqrt{a^2 - x^2}$	$x = a \sin \theta$	$1 - \sin^2 \theta = \cos^2 \theta$

Integrand form	Substitution	Identity used
$\sqrt{a^2 + x^2}$	$x = a \tan \theta$	$1 + \tan^2 \theta = \sec^2 \theta$
$\sqrt{x^2 - a^2}$	$x = a \sec \theta$	$\sec^2 \theta - 1 = \tan^2 \theta$

Procedure: Substitute, simplify the radical (take positive square root), integrate in θ , back-substitute using a reference triangle.

Example: $\int \frac{\sqrt{9-x^2}}{x^2} dx$

$$x = 3 \sin \theta, dx = 3 \cos \theta d\theta, \sqrt{9-x^2} = 3 \cos \theta$$

$$= \int \frac{3 \cos \theta}{9 \sin^2 \theta} \cdot 3 \cos \theta d\theta = \int \frac{\cos^2 \theta}{\sin^2 \theta} d\theta = \int \cot^2 \theta d\theta = \int (\csc^2 \theta - 1) d\theta$$

$$= -\cot \theta - \theta + C$$

$$\text{Back-sub: } \sin \theta = x/3, \text{ so } \cos \theta = \sqrt{9-x^2}/3, \cot \theta = \sqrt{9-x^2}/x, \theta = \arcsin(x/3)$$

$$= -\frac{\sqrt{9-x^2}}{x} - \arcsin \frac{x}{3} + C$$

Completing the square first: $\int \frac{dx}{\sqrt{x^2+4x+5}} = \int \frac{dx}{\sqrt{(x+2)^2+1}}$, then $u = x + 2$, trig sub $u = \tan \theta$.

Common Mistakes:

- Forgetting dx in terms of $d\theta$ after substitution
- Taking the wrong sign of the square root ($\sqrt{a^2 \cos^2 \theta} = a|\cos \theta|$, assume $\cos \theta > 0$ for $\theta \in (-\pi/2, \pi/2)$)
- Forgetting to back-substitute at the end

3.3 Improper Integrals

Type I (infinite limits):

$$\int_a^\infty f(x) dx = \lim_{t \rightarrow \infty} \int_a^t f(x) dx$$

Type II (discontinuous integrand at $c \in [a, b]$):

$$\int_a^b f(x) dx = \lim_{t \rightarrow c^-} \int_a^t f(x) dx + \lim_{t \rightarrow c^+} \int_t^b f(x) dx$$

p-integral results:

$$\int_1^\infty \frac{1}{x^p} dx \text{ converges iff } p > 1, \text{ value} = \frac{1}{p-1}$$

$$\int_0^1 \frac{1}{x^p} dx \text{ converges iff } p < 1$$

Example: $\int_1^\infty \frac{1}{x^2} dx = \lim_{t \rightarrow \infty} \left[-\frac{1}{x}\right]_1^t = \lim_{t \rightarrow \infty} \left(-\frac{1}{t} + 1\right) = 1$

Comparison Test: If $0 \leq f(x) \leq g(x)$ for $x \geq a$:

- $\int g$ converges $\Rightarrow \int f$ converges
- $\int f$ diverges $\Rightarrow \int g$ diverges

Common Mistakes:

- Evaluating $\int_{-1}^1 \frac{1}{x} dx$ directly without splitting at $x = 0$ (the integrand is discontinuous there — the integral diverges)

3.4 Applications of Integration

Area Between Curves

$$A = \int_a^b [f(x) - g(x)] dx \quad \text{where } f(x) \geq g(x) \text{ on } [a, b]$$

If curves cross: split at intersection points.

Volumes — Disk/Washer Method (rotation about horizontal axis)

$$V = \pi \int_a^b [R(x)]^2 dx \quad (\text{disk})$$

$$V = \pi \int_a^b ([R(x)]^2 - [r(x)]^2) dx \quad (\text{washer, outer radius } R, \text{ inner radius } r)$$

Volumes — Shell Method (rotation about vertical axis)

$$V = 2\pi \int_a^b x \cdot f(x) dx$$

When to use: Shell is often easier when washers would require solving for x in terms of y .

Arc Length

$$L = \int_a^b \sqrt{1 + [f'(x)]^2} dx$$

Surface Area of Revolution (about x -axis)

$$S = 2\pi \int_a^b f(x) \sqrt{1 + [f'(x)]^2} dx$$

Work

$$W = \int_a^b F(x) dx \quad (\text{force varying with position})$$

Common Mistakes:

- Washer method: squaring $(R - r)$ instead of $R^2 - r^2$
- Shell method: forgetting the factor of 2π and the x factor

3.5 Sequences and Series

Sequences

A sequence $\{a_n\}$ converges if $\lim_{n \rightarrow \infty} a_n = L$ (finite). Otherwise it diverges.

Apply L'Hôpital by treating n as continuous: $\lim_{n \rightarrow \infty} \frac{\ln n}{n} = 0$.

Geometric Series

$$\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r} \quad \text{iff } |r| < 1$$

Diverges if $|r| \geq 1$.

Convergence Tests

Test	Statement	Best For
Divergence Test	If $\lim a_n \neq 0$, diverges	Quick check; cannot confirm convergence
Integral Test	$\sum a_n$ and $\int f(x) dx$ both converge or both diverge (f positive, decreasing, $a_n = f(n)$)	$a_n = 1/n^p$, etc.
p-Series	$\sum 1/n^p$ converges iff $p > 1$	
Direct Comparison	$0 \leq a_n \leq b_n$: if $\sum b_n$ converges, so does $\sum a_n$	Known comparison series
Limit Comparison	If $\lim a_n/b_n = c > 0$, both series behave the same	Rational-like terms
Alternating Series Test	$\sum (-1)^n b_n$ converges if $b_n \geq 0$, decreasing, $b_n \rightarrow 0$	Alternating series
Ratio Test	$L = \lim$	a_{n+1}/a_n
Root Test	$L = \lim$	a_n

Absolute vs. conditional convergence:

- Absolutely convergent: $\sum |a_n|$ converges (implies convergence)
- Conditionally convergent: $\sum a_n$ converges but $\sum |a_n|$ diverges (e.g., alternating harmonic series)

Power Series

$$\sum_{n=0}^{\infty} c_n (x-a)^n$$

Radius of convergence via ratio test: $\frac{1}{R} = \lim \left| \frac{c_{n+1}}{c_n} \right|$

After finding R : check each endpoint $x = a \pm R$ separately (ratio test is inconclusive there).

Taylor and Maclaurin Series

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n$$

Maclaurin series to memorize:

Function	Series	Converges for
e^x	$\sum_{n=0}^{\infty} \frac{x^n}{n!}$	all x
$\sin x$	$\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$	all x
$\cos x$	$\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$	all x
$\ln(1+x)$	$\sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n}$	$-1 < x \leq 1$
$\frac{1}{1-x}$	$\sum_{n=0}^{\infty} x^n$	\$
$\arctan x$	$\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1}$	\$

Operations: differentiate or integrate term by term within the radius of convergence. Substitute directly (e.g., e^{-x^2} from e^x).

3.6 Parametric and Polar

Parametric Curves

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}, \quad \frac{d^2y}{dx^2} = \frac{d(dy/dx)/dt}{dx/dt}$$

Arc length: $L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$

Polar Coordinates

$$x = r \cos \theta, y = r \sin \theta, r^2 = x^2 + y^2, \tan \theta = y/x$$

Common curves:

- Circle: $r = a$ (radius a) or $r = 2a \cos \theta$ (center on x -axis)
- Cardioid: $r = a(1 + \cos \theta)$
- Rose: $r = a \cos(n\theta)$ — n petals if n odd, $2n$ if n even
- Lemniscate: $r^2 = a^2 \cos(2\theta)$

Area in polar:

$$A = \frac{1}{2} \int_{\alpha}^{\beta} r^2 d\theta$$

For area between curves: $A = \frac{1}{2} \int_{\alpha}^{\beta} (r_{\text{outer}}^2 - r_{\text{inner}}^2) d\theta$

Arc length in polar:

$$L = \int_{\alpha}^{\beta} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

Common Mistakes:

- Series: applying the ratio test to get $L = 1$ and concluding convergence (inconclusive!)
- Polar area: using $\int r d\theta$ instead of $\frac{1}{2} \int r^2 d\theta$

SUBJECT 4: CALCULUS 3

4.1 Vectors and 3D Geometry

Vector Operations

For $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$, $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$:

Dot product: $\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3 = |\mathbf{a}||\mathbf{b}| \cos \theta$

Projection of $\mathbf{b}$ onto $\mathbf{a}$:

$$\text{comp}_{\mathbf{a}} \mathbf{b} = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}|}, \quad \text{proj}_{\mathbf{a}} \mathbf{b} = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}|^2} \mathbf{a}$$

Cross product:

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \langle a_2b_3 - a_3b_2, a_3b_1 - a_1b_3, a_1b_2 - a_2b_1 \rangle$$

- $|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}||\mathbf{b}| \sin \theta$ = area of parallelogram
- $\mathbf{a} \times \mathbf{b}$ is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$ (right-hand rule)
- $\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$ (anti-commutative)

Lines in 3D

Through point $P_0 = (x_0, y_0, z_0)$ with direction $\mathbf{d} = \langle a, b, c \rangle$:

Parametric: $x = x_0 + at, y = y_0 + bt, z = z_0 + ct$

Symmetric: $\frac{x-x_0}{a} = \frac{y-y_0}{b} = \frac{z-z_0}{c}$

Planes

Through P_0 with normal $\mathbf{n} = \langle a, b, c \rangle$:

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

Distance from point (x_1, y_1, z_1) to plane $ax + by + cz = d$:

$$D = \frac{|ax_1 + by_1 + cz_1 - d|}{\sqrt{a^2 + b^2 + c^2}}$$

Quadric Surfaces (Standard Forms)

Surface	Equation
Ellipsoid	$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$
Elliptic paraboloid	$z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$
Hyperbolic paraboloid (saddle)	$z = \frac{x^2}{a^2} - \frac{y^2}{b^2}$
Cone	$z^2 = \frac{x^2}{a^2} + \frac{y^2}{b^2}$
Hyperboloid of one sheet	$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$

Common Mistakes:

- Cross product: computing $\mathbf{a} \times \mathbf{b}$ when you need $\mathbf{b} \times \mathbf{a}$ (sign error in normal direction)
- Dot product: getting the angle wrong because $\cos^{-1}$ gives $[0, \pi]$ only

4.2 Vector-Valued Functions

$$\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle$$

$$\mathbf{r}'(t) = \langle f'(t), g'(t), h'(t) \rangle, \quad \int \mathbf{r}(t) dt = \left\langle \int f dt, \int g dt, \int h dt \right\rangle + \mathbf{C}$$

Arc length: $L = \int_a^b |\mathbf{r}'(t)| dt$

Unit tangent: $\mathbf{T} = \frac{\mathbf{r}'}{|\mathbf{r}'|}$

Curvature: $\kappa = \frac{|\mathbf{T}'|}{|\mathbf{r}'|} = \frac{|\mathbf{r}' \times \mathbf{r}''|}{|\mathbf{r}'|^3}$

Principal normal: $\mathbf{N} = \frac{\mathbf{T}'}{|\mathbf{T}'|}$, **Binormal:** $\mathbf{B} = \mathbf{T} \times \mathbf{N}$

Acceleration components:

$$a_T = \frac{d|\mathbf{v}|}{dt} = \frac{\mathbf{r}' \cdot \mathbf{r}''}{|\mathbf{r}'|}, \quad a_N = \kappa |\mathbf{v}|^2 = \frac{|\mathbf{r}' \times \mathbf{r}''|}{|\mathbf{r}'|}$$

4.3 Partial Derivatives

$$f_x = \frac{\partial f}{\partial x} : \text{differentiate w.r.t. } x, \text{ treating } y \text{ as constant}$$

Clairaut's theorem: If f_{xy} and f_{yx} are both continuous, then $f_{xy} = f_{yx}$.

Chain Rule (Multivariable)

If $z = f(x, y)$ and $x = g(t)$, $y = h(t)$:

$$\frac{dz}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}$$

If $z = f(x, y)$ and $x = g(s, t)$, $y = h(s, t)$:

$$\frac{\partial z}{\partial s} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial s}$$

Implicit differentiation: For $F(x, y) = 0$:

$$\frac{dy}{dx} = -\frac{F_x}{F_y}$$

For $F(x, y, z) = 0$: $\frac{\partial z}{\partial x} = -\frac{F_x}{F_z}$

Gradient

$$\nabla f = \langle f_x, f_y, f_z \rangle$$

- Points in direction of steepest increase; magnitude = rate of steepest ascent
- Perpendicular to level curves (2D) and level surfaces (3D)

Directional derivative: $D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$ ($\mathbf{u}$ must be a unit vector)

Tangent plane to $z = f(x, y)$ at (x_0, y_0, z_0) :

$$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

4.4 Multivariable Optimization

Critical points: Solve $\nabla f = \mathbf{0}$, i.e., $f_x = 0$ and $f_y = 0$ simultaneously.

Second derivative test: Compute $D = f_{xx}f_{yy} - (f_{xy})^2$ at each critical point.

D	f_{xx}	Conclusion
$D > 0$	> 0	Local minimum
$D > 0$	< 0	Local maximum
$D < 0$	—	Saddle point
$D = 0$	—	Inconclusive

Lagrange Multipliers

Optimize $f(x, y, z)$ subject to $g(x, y, z) = k$:

$$\nabla f = \lambda \nabla g, \quad g = k$$

This gives a system of equations; solve for x, y, z, λ .

Two constraints: $\nabla f = \lambda \nabla g + \mu \nabla h$

Example: Maximize $f(x, y) = xy$ subject to $x + y = 10$.

$$\nabla f = \langle y, x \rangle = \lambda \langle 1, 1 \rangle = \lambda \nabla(x + y)$$

$y = \lambda, x = \lambda$, so $x = y$. Combined with $x + y = 10$: $x = y = 5$. Max = 25.

Common Mistakes:

- Forgetting to check the boundary when finding absolute extrema on a closed region
- SDT: computing $D = f_{xx}f_{yy} - f_{xy}$ instead of $f_{xx}f_{yy} - (f_{xy})^2$

4.5 Multiple Integrals

Double Integrals — Setup

$$\iint_R f(x, y) dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx$$

Switching order: Sketch the region R , re-describe its bounds with the other variable as the outer integral.

Polar: $\iint_R f(x, y) dA = \int_\alpha^\beta \int_{r_1(\theta)}^{r_2(\theta)} f(r \cos \theta, r \sin \theta) \cdot r dr d\theta$

Note: $dA = r dr d\theta$ (don't forget the Jacobian r !)

Triple Integrals

Cylindrical coordinates: $x = r \cos \theta, y = r \sin \theta, z = z; dV = r dr d\theta dz$

Spherical coordinates: $x = \rho \sin \phi \cos \theta, y = \rho \sin \phi \sin \theta, z = \rho \cos \phi; dV = \rho^2 \sin \phi d\rho d\phi d\theta$

- ρ : distance from origin
- ϕ : angle from positive z -axis (polar angle), $\phi \in [0, \pi]$
- θ : azimuthal angle from x -axis, $\theta \in [0, 2\pi]$

When to use: Cylindrical for cylinders/cones; spherical for spheres and cones from origin.

Change of Variables (Jacobian)

$$\iint_R f(x, y) dA = \iint_S f(x(u, v), y(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv$$

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix}$$

4.6 Vector Calculus

Line Integrals

Scalar: $\int_C f ds = \int_a^b f(\mathbf{r}(t)) |\mathbf{r}'(t)| dt$

Vector (work): $\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt$

Conservative Fields and Potential Functions

$\mathbf{F} = \langle P, Q, R \rangle$ is conservative on a simply connected domain iff $\nabla \times \mathbf{F} = \mathbf{0}$ (i.e., $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$, etc.)

Fundamental Theorem for Line Integrals:

$$\int_C \nabla f \cdot d\mathbf{r} = f(\mathbf{r}(b)) - f(\mathbf{r}(a))$$

Finding potential f : Integrate P w.r.t. x , then differentiate w.r.t. y and match Q .

Curl and Divergence

$$\nabla \times \mathbf{F} = \text{curl } \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial_x & \partial_y & \partial_z \\ P & Q & R \end{vmatrix}$$

$$\nabla \cdot \mathbf{F} = \text{div } \mathbf{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$$

The Four Big Theorems

Green's Theorem (2D, relates line integral around closed curve C to double integral over enclosed region D):

$$\oint_C P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

Stokes' Theorem (generalizes Green's to 3D surface S with boundary C):

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$$

Divergence Theorem (relates flux through closed surface S to volume integral over enclosed E):

$$\oiint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_E (\nabla \cdot \mathbf{F}) dV$$

When to use each:

- Green's: 2D closed curve, hard to integrate directly
- Stokes': 3D closed curve, or simplify a surface integral to a different surface
- Divergence: closed surface, outward flux, often easier than direct surface integration

Common Mistakes:

- Spherical: mixing up ϕ (from z -axis) and θ (azimuthal), and forgetting $\sin \phi$ in dV
- Green's: wrong sign ($Q_x - P_y$ not $P_x - Q_y$)
- Divergence theorem: applying to non-closed surfaces

SUBJECT 5: DIFFERENTIAL EQUATIONS

5.1 First-Order ODEs

Separable Equations

Form: $\frac{dy}{dx} = g(x)h(y)$

Procedure: $\frac{dy}{h(y)} = g(x) dx$, then integrate both sides.

Example: $\frac{dy}{dx} = xy, y(0) = 2$

$$\frac{dy}{y} = x dx \Rightarrow \ln |y| = \frac{x^2}{2} + C_1 \Rightarrow y = Ce^{x^2/2}$$

$$y(0) = 2 \Rightarrow C = 2 \Rightarrow y = 2e^{x^2/2}$$

Linear First-Order: $\frac{dy}{dx} + P(x)y = Q(x)$

Integrating factor: $\mu(x) = e^{\int P(x) dx}$

General solution: $y = \frac{1}{\mu(x)} \int \mu(x)Q(x) dx + \frac{C}{\mu(x)}$

Equivalently: multiply both sides by μ ; left side becomes $\frac{d}{dx}[\mu y]$; integrate.

Example: $y' + \frac{2}{x}y = 4x, y(1) = 2$

$$\mu = e^{\int 2/x dx} = e^{2 \ln x} = x^2$$

$$\frac{d}{dx}[x^2 y] = 4x \cdot x^2 = 4x^3$$

$$x^2 y = x^4 + C \Rightarrow y = x^2 + Cx^{-2}$$

$$y(1) = 1 + C = 2 \Rightarrow C = 1 \Rightarrow y = x^2 + x^{-2}$$

Exact Equations

Form: $M(x, y) dx + N(x, y) dy = 0$

Test for exactness: $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

Procedure: Find $F(x, y)$ such that $F_x = M$ and $F_y = N$:

1. $F = \int M dx + g(y)$
2. Differentiate w.r.t. y : $F_y = (\int M dx)_y + g'(y) = N$
3. Solve for $g'(y)$, integrate for $g(y)$
4. Solution: $F(x, y) = C$

Example: $(2xy - y^2) dx + (x^2 - 2xy) dy = 0$

$$M_y = 2x - 2y = N_x \checkmark$$

$$F = \int (2xy - y^2) dx = x^2 y - xy^2 + g(y)$$

$$F_y = x^2 - 2xy + g'(y) = x^2 - 2xy \Rightarrow g'(y) = 0 \Rightarrow g = C_1$$

$$\text{Solution: } x^2 y - xy^2 = C$$

Bernoulli Equations

Form: $y' + P(x)y = Q(x)y^n$ (nonlinear when $n \neq 0, 1$)

Substitution: $v = y^{1-n}$ transforms to linear ODE:

$$v' + (1 - n)P(x)v = (1 - n)Q(x)$$

Homogeneous Equations (Type: $dy/dx = f(y/x)$)

Substitution: $v = y/x$ (so $y = vx$, $y' = v + xv'$) $\rightarrow$ separable in v and x .

Existence and Uniqueness

For $y' = f(x, y)$, $y(x_0) = y_0$: if f and $\partial f / \partial y$ are continuous near (x_0, y_0) , a unique solution exists in some interval around x_0 .

Common Mistakes:

- Linear ODE: computing μ incorrectly from $P(x)$ (make sure equation is in standard form first)
- Exact: computing $g'(y)$ and finding x terms remain (not exact, or computational error)

5.2 Second-Order Linear ODEs with Constant Coefficients

Homogeneous: $ay'' + by' + cy = 0$

Characteristic equation: $ar^2 + br + c = 0$, $r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$\Delta = b^2 - 4ac$	Roots	General solution y_h
$\Delta > 0$	Two distinct real r_1, r_2	$C_1 e^{r_1 x} + C_2 e^{r_2 x}$
$\Delta = 0$	Repeated real root r	$(C_1 + C_2 x) e^{rx}$
$\Delta < 0$	Complex $\alpha \pm \beta i$	$e^{\alpha x} (C_1 \cos \beta x + C_2 \sin \beta x)$

Example: $y'' - 5y' + 6y = 0$

$$r^2 - 5r + 6 = 0 \Rightarrow (r - 2)(r - 3) = 0 \Rightarrow r = 2, 3$$

$$y = C_1 e^{2x} + C_2 e^{3x}$$

Example (complex): $y'' + 4y = 0$

$$r^2 + 4 = 0 \Rightarrow r = \pm 2i \quad (\alpha = 0, \beta = 2)$$

$$y = C_1 \cos 2x + C_2 \sin 2x$$

Non-Homogeneous: $ay'' + by' + cy = g(x)$

General solution: $y = y_h + y_p$

Method of Undetermined Coefficients (guess table for y_p):

$g(x)$ form	Guess for y_p
$P_n(x)$ (degree n poly)	$x^s (A_n x^n + \dots + A_0)$
$e^{\alpha x}$	$x^s A e^{\alpha x}$

$g(x)$ form	Guess for y_p
$\cos(\beta x)$ or $\sin(\beta x)$	$x^s(A \cos \beta x + B \sin \beta x)$
$e^{\alpha x} \cos(\beta x)$ or $e^{\alpha x} \sin(\beta x)$	$x^s e^{\alpha x}(A \cos \beta x + B \sin \beta x)$
$P_n(x)e^{\alpha x}$	$x^s e^{\alpha x}(A_n x^n + \dots + A_0)$

Modification rule: s = smallest non-negative integer such that no term in y_p duplicates y_h . Usually $s = 0$; use $s = 1$ if the exponential matches a root, $s = 2$ if it's a repeated root.

Example: $y'' - 3y' + 2y = 4e^{2x}$

$$y_h: r^2 - 3r + 2 = 0, (r - 1)(r - 2) = 0, y_h = C_1 e^x + C_2 e^{2x}$$

Since e^{2x} appears in y_h : guess $y_p = A x e^{2x}$ (use $s = 1$)

$$y_p' = A e^{2x}(1 + 2x), y_p'' = A e^{2x}(4 + 4x)$$

$$\text{Substitute: } A e^{2x}(4 + 4x) - 3A e^{2x}(1 + 2x) + 2A x e^{2x} = 4e^{2x}$$

$$A e^{2x}(4 + 4x - 3 - 6x + 2x) = A e^{2x} = 4e^{2x} \Rightarrow A = 4$$

$$y_p = 4x e^{2x}$$

Variation of Parameters (general — works for any $g(x)$)

Given $y_h = C_1 y_1 + C_2 y_2$:

$$\text{Wronskian: } W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_2 y_1'$$

$$u_1' = \frac{-y_2 g}{aW}, \quad u_2' = \frac{y_1 g}{aW}$$

Integrate to get u_1, u_2 . Then $y_p = u_1 y_1 + u_2 y_2$.

Reduction of Order

If y_1 is a known solution, let $y_2 = v(x)y_1$, substitute into the ODE, and solve the resulting first-order ODE in v' .

Common Mistakes:

- UDC: not checking if y_p guess duplicates y_h before differentiating
- Variation of parameters: using $g(x)$ directly without putting the ODE in standard form first (divide by a so coefficient of y'' is 1)

5.3 Laplace Transforms

Definition

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t) dt$$

Complete Transform Table

$f(t)$	$F(s) = \mathcal{L}\{f\}$	Notes
1	$1/s$	$s > 0$
t^n	$n!/s^{n+1}$	$s > 0$
e^{at}	$1/(s-a)$	$s > a$
$t^n e^{at}$	$n!/(s-a)^{n+1}$	
$\sin(bt)$	$b/(s^2 + b^2)$	
$\cos(bt)$	$s/(s^2 + b^2)$	
$e^{at} \sin(bt)$	$b/((s-a)^2 + b^2)$	
$e^{at} \cos(bt)$	$(s-a)/((s-a)^2 + b^2)$	
$\sinh(bt)$	$b/(s^2 - b^2)$	
$\cosh(bt)$	$s/(s^2 - b^2)$	
$u(t-a)$	e^{-as}/s	unit step
$\delta(t-a)$	e^{-as}	Dirac delta
$f(t-a)u(t-a)$	$e^{-as}F(s)$	2nd shift theorem
$e^{at}f(t)$	$F(s-a)$	1st shift theorem
$f'(t)$	$sF(s) - f(0)$	
$f''(t)$	$s^2F(s) - sf(0) - f'(0)$	
$f^{(n)}(t)$	$s^nF(s) - s^{n-1}f(0) - \dots - f^{(n-1)}(0)$	
$(f * g)(t) = \int_0^t f(\tau)g(t-\tau)d\tau$	$F(s)G(s)$	convolution

Solving IVPs with Laplace

Procedure:

1. Take $\mathcal{L}$ of both sides
2. Apply derivative property to express $\mathcal{L}\{y'\}$ and $\mathcal{L}\{y''\}$ in terms of $Y(s) = \mathcal{L}\{y\}$ and ICs
3. Solve algebraically for $Y(s)$
4. Apply partial fractions and inverse Laplace to find $y(t)$

Example: $y'' + 3y' + 2y = e^{-t}$, $y(0) = 0$, $y'(0) = 1$

Take Laplace: $[s^2Y - s \cdot 0 - 1] + 3[sY - 0] + 2Y = \frac{1}{s+1}$

$$(s^2 + 3s + 2)Y = 1 + \frac{1}{s+1} = \frac{s+2}{s+1}$$

$$Y = \frac{s+2}{(s+1)(s+1)(s+2)} = \frac{1}{(s+1)^2}$$

Wait — let me redo: $(s+1)(s+2)Y = \frac{s+2}{s+1}$, so $Y = \frac{1}{(s+1)^2}$

$$\mathcal{L}^{-1}\left\{\frac{1}{(s+1)^2}\right\} = te^{-t}$$

Inverse Laplace — Key Techniques

1. **Partial fractions:** decompose $Y(s)$ into recognizable forms from the table
2. **Complete the square:** for complex poles, get $(s - a)^2 + b^2$ in denominator
3. **First shift theorem:** $\mathcal{L}^{-1}\{F(s - a)\} = e^{at}f(t)$

Unit Step and Discontinuous Forcing

$$u(t - a) = \begin{cases} 0 & t < a \\ 1 & t \geq a \end{cases}$$

$$\mathcal{L}\{u(t - a)g(t)\} = e^{-as}\mathcal{L}\{g(t + a)\}$$

Common Mistakes:

- Applying the 1st shift theorem backwards: $F(s - a)$ in s -domain corresponds to e^{at} (positive exponent) in time domain
- Forgetting initial condition terms in $\mathcal{L}\{y''\}$

5.4 Systems of First-Order Linear ODEs

Matrix Form

Convert to $\mathbf{x}' = A\mathbf{x}$ where $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$

Converting a 2nd-order ODE: For $y'' + py' + qy = 0$, let $x_1 = y$, $x_2 = y'$:

$$\begin{pmatrix} x_1' \\ x_2' \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -q & -p \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

Solution via Eigenvalues

Find eigenvalues from $\det(A - \lambda I) = 0$.

Case 1 — Distinct real eigenvalues λ_1, λ_2 with eigenvectors $\mathbf{v}_1, \mathbf{v}_2$:

$$\mathbf{x} = C_1\mathbf{v}_1e^{\lambda_1 t} + C_2\mathbf{v}_2e^{\lambda_2 t}$$

Case 2 — Repeated eigenvalue λ (algebraic multiplicity 2):

If only one eigenvector $\mathbf{v}_1$ exists, find generalized eigenvector $\mathbf{v}_2$: $(A - \lambda I)\mathbf{v}_2 = \mathbf{v}_1$

$$\mathbf{x} = C_1\mathbf{v}_1e^{\lambda t} + C_2(\mathbf{v}_1 t + \mathbf{v}_2)e^{\lambda t}$$

Case 3 — Complex eigenvalues $\alpha \pm \beta i$ with eigenvector $\mathbf{v} = \mathbf{a} + i\mathbf{b}$:

$$\mathbf{x} = C_1e^{\alpha t}(\mathbf{a} \cos \beta t - \mathbf{b} \sin \beta t) + C_2e^{\alpha t}(\mathbf{a} \sin \beta t + \mathbf{b} \cos \beta t)$$

Example: $\mathbf{x}' = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix} \mathbf{x}$

$$\det(A - \lambda I) = (1 - \lambda)(2 - \lambda) - 6 = \lambda^2 - 3\lambda - 4 = (\lambda - 4)(\lambda + 1) = 0$$

$$\lambda_1 = 4: (A - 4I)\mathbf{v} = 0 \Rightarrow \mathbf{v}_1 = \langle 2, 3 \rangle^T$$

$$\lambda_2 = -1: (A + I)\mathbf{v} = 0 \Rightarrow \mathbf{v}_2 = \langle -1, 1 \rangle^T$$

$$\mathbf{x} = C_1 \begin{pmatrix} 2 \\ 3 \end{pmatrix} e^{4t} + C_2 \begin{pmatrix} -1 \\ 1 \end{pmatrix} e^{-t}$$

5.5 Numerical Methods for ODEs

Euler's Method

$$y_{n+1} = y_n + h \cdot f(x_n, y_n)$$

h = step size. Global error $\sim O(h)$.

Improved Euler / Heun's Method

$$k_1 = f(x_n, y_n), \quad k_2 = f(x_n + h, y_n + hk_1)$$

$$y_{n+1} = y_n + \frac{h}{2}(k_1 + k_2)$$

Global error $\sim O(h^2)$.

Runge-Kutta 4th Order (RK4)

$$k_1 = f(x_n, y_n)$$

$$k_2 = f(x_n + h/2, y_n + hk_1/2)$$

$$k_3 = f(x_n + h/2, y_n + hk_2/2)$$

$$k_4 = f(x_n + h, y_n + hk_3)$$

$$y_{n+1} = y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

Global error $\sim O(h^4)$. Standard choice for most problems.

5.6 Qualitative Analysis

Autonomous Equations: $\frac{dy}{dt} = f(y)$

Equilibrium solutions: values y^* where $f(y^*) = 0$

Stability:

- **Stable (attractor):** $f'(y^*) < 0$ — nearby solutions converge
- **Unstable (repeller):** $f'(y^*) > 0$ — nearby solutions diverge
- **Semi-stable:** f doesn't change sign at y^*

Phase line: plot $f(y)$ vs. y ; where $f > 0$, solutions increase; where $f < 0$, solutions decrease.

Example: $dy/dt = y(1-y)(2-y)$

Equilibria: $y = 0, 1, 2$. Check signs of f in intervals:

- $y < 0$: $(-)(+)(+) < 0 \rightarrow$ decreasing
- $0 < y < 1$: $(+)(+)(-) < 0 \rightarrow$ wait: $(+)(1-y > 0)(2-y > 0) > 0 \rightarrow$ increasing
- $1 < y < 2$: $(+)(-)(+) < 0 \rightarrow$ decreasing
- $y > 2$: $(+)(-)(+) < 0 \rightarrow$ wait: $(+)(\text{neg})(\text{pos}) < 0 \rightarrow$ no: check sign carefully. $y > 2$: $y(+)$, $1-y(-)$, $2-y(-)$: $(+)(-)(-) > 0 \rightarrow$ increasing.

Result: $y = 0$ unstable, $y = 1$ stable, $y = 2$ unstable.

SUBJECT 6: LINEAR ALGEBRA

6.1 Vectors and Vector Spaces

Vector Space Axioms

A set V with operations $+$ and scalar multiplication satisfies all 10 axioms (closure, commutativity, associativity, identity, inverse, distributivity, etc.). Key examples: $\mathbb{R}^n$, P_n (polynomials of degree $\leq n$), $M_{m \times n}$ (matrices).

Subspace Test

$H \subseteq V$ is a subspace iff:

1. $\mathbf{0} \in H$
2. Closed under addition: $\mathbf{u}, \mathbf{v} \in H \Rightarrow \mathbf{u} + \mathbf{v} \in H$
3. Closed under scalar multiplication: $\mathbf{u} \in H, c \in \mathbb{R} \Rightarrow c\mathbf{u} \in H$

Linear Independence

$\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ is linearly independent iff $c_1\mathbf{v}_1 + \dots + c_p\mathbf{v}_p = \mathbf{0}$ implies $c_1 = \dots = c_p = 0$.

Test: set up the matrix $[\mathbf{v}_1 \cdots \mathbf{v}_p]$ and row reduce. Independent iff no free variables.

Basis and Dimension

A **basis** for H is a linearly independent set that spans H . The **dimension** is the number of vectors in any basis.

Rank-Nullity Theorem

For $m \times n$ matrix A :

$$\text{rank}(A) + \text{nullity}(A) = n$$

where $\text{rank} = \dim(\text{col space}) = \text{number of pivot columns}$, $\text{nullity} = \dim(\text{null space}) = \text{number of free variables}$.

6.2 Matrices and Operations

Matrix Multiplication

$$(AB)_{ij} = \sum_k A_{ik} B_{kj} \text{ (row of } A \text{ dotted with column of } B)$$

AB is defined only if columns of A = rows of B . $AB \neq BA$ in general.

Properties:

- $(AB)C = A(BC)$ (associative)
- $A(B + C) = AB + AC$ (distributive)
- $(AB)^T = B^T A^T$
- $(AB)^{-1} = B^{-1} A^{-1}$

Matrix Inverse

A^{-1} exists iff $\det(A) \neq 0$ iff A has n pivot positions.

$$\mathbf{2 \times 2 \text{ formula:}} \quad A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \Rightarrow A^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

General method: Row reduce $[A|I] \rightarrow [I|A^{-1}]$

6.3 Systems of Linear Equations

Gaussian Elimination to REF

Row operations (preserve solution set):

1. Swap two rows
2. Multiply a row by a nonzero scalar
3. Add a multiple of one row to another

REF: each leading entry is to the right of all leading entries above it.

RREF: each leading entry is 1, and it's the only nonzero entry in its column.

Solution structure: number of solutions determined by consistency and free variables.

- Inconsistent (row $[0 \ \cdots \ 0 \mid b], b \neq 0$): no solution
- Consistent, no free variables: exactly one solution
- Consistent, k free variables: infinitely many solutions (parametric form, k -dimensional solution set)

Example: Solve
$$\begin{cases} x + 2y + z = 4 \\ 2x + 5y + 3z = 9 \\ x + 3y + 2z = 5 \end{cases}$$

Augmented matrix row reduction:

$$\begin{pmatrix} 1 & 2 & 1 & 4 \\ 2 & 5 & 3 & 9 \\ 1 & 3 & 2 & 5 \end{pmatrix} \xrightarrow{R_2-2R_1, R_3-R_1} \begin{pmatrix} 1 & 2 & 1 & 4 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix} \xrightarrow{R_3-R_2} \begin{pmatrix} 1 & 2 & 1 & 4 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Free variable: $z = t$. Back-substitute: $y = 1 - t$, $x = 4 - 2(1 - t) - t = 2 + t$.

Solution: $(x, y, z) = (2, 1, 0) + t(1, -1, 1)$.

6.4 Determinants

Cofactor Expansion (along row i)

$$\det(A) = \sum_{j=1}^n (-1)^{i+j} a_{ij} M_{ij}$$

where M_{ij} is the minor (det of the $(n-1) \times (n-1)$ submatrix with row i and column j removed).

Sign pattern: $\begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}$

Example: $\det \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$, expand along row 1:

$$\begin{aligned} &= 1 \det \begin{pmatrix} 5 & 6 \\ 8 & 9 \end{pmatrix} - 2 \det \begin{pmatrix} 4 & 6 \\ 7 & 9 \end{pmatrix} + 3 \det \begin{pmatrix} 4 & 5 \\ 7 & 8 \end{pmatrix} \\ &= 1(45 - 48) - 2(36 - 42) + 3(32 - 35) = -3 + 12 - 9 = 0 \text{ (rows are linearly dependent)} \end{aligned}$$

Properties

- $\det(AB) = \det(A) \det(B)$
- $\det(A^T) = \det(A)$
- $\det(A^{-1}) = 1/\det(A)$
- Swapping two rows multiplies det by -1
- Multiplying a row by k multiplies det by k
- Adding a multiple of one row to another: det unchanged
- $\det(A) = 0 \Leftrightarrow A$ is singular (not invertible)

Cramer's Rule

For $n \times n$ system $A\mathbf{x} = \mathbf{b}$ with $\det(A) \neq 0$:

$$x_i = \frac{\det(A_i)}{\det(A)}$$

where A_i is A with column i replaced by $\mathbf{b}$.

Common Mistakes:

- Getting the sign pattern wrong in cofactor expansion
- $\det(kA) = k^n \det(A)$ (not $k \det(A)$)

6.5 Eigenvalues and Eigenvectors

Definition and Procedure

λ is an eigenvalue of A and $\mathbf{v} \neq \mathbf{0}$ is the corresponding eigenvector iff $A\mathbf{v} = \lambda\mathbf{v}$, equivalently $(A - \lambda I)\mathbf{v} = \mathbf{0}$.

Step 1: Solve $\det(A - \lambda I) = 0$ (characteristic equation) for λ .

Step 2: For each λ , solve $(A - \lambda I)\mathbf{v} = \mathbf{0}$ by row reduction. Each free variable gives one eigenvector.

Example: $A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$

$$\det(A - \lambda I) = (3 - \lambda)^2 - 1 = \lambda^2 - 6\lambda + 8 = (\lambda - 4)(\lambda - 2)$$

$$\lambda_1 = 4: (A - 4I) = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \rightarrow \mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\lambda_2 = 2: (A - 2I) = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \rightarrow \mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

Algebraic vs. Geometric Multiplicity

- **Algebraic multiplicity:** multiplicity of λ as a root of the characteristic polynomial
- **Geometric multiplicity:** $\dim(\text{null}(A - \lambda I))$ = number of linearly independent eigenvectors for λ
- Always: geometric $\leq$ algebraic

Diagonalization

$A = PDP^{-1}$ where D is diagonal (eigenvalues on diagonal) and P has eigenvectors as columns.

Condition: A is diagonalizable iff it has n linearly independent eigenvectors (equivalently, geometric multiplicity = algebraic multiplicity for each eigenvalue).

Powers: $A^k = PD^kP^{-1}$ where D^k has λ_i^k on the diagonal.

Symmetric matrices: always diagonalizable; eigenvectors for distinct eigenvalues are orthogonal; can diagonalize as $A = QDQ^T$ where Q is orthogonal ($Q^{-1} = Q^T$).

Common Mistakes:

- Putting eigenvectors in the wrong order in P — they must match the order of eigenvalues in D
- A repeated eigenvalue does not automatically mean the matrix is not diagonalizable — check geometric multiplicity

6.6 Orthogonality

Orthogonal and Orthonormal Sets

$\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ is orthogonal if $\mathbf{u}_i \cdot \mathbf{u}_j = 0$ for $i \neq j$.

Orthonormal: additionally $|\mathbf{u}_i| = 1$ for all i .

An orthogonal set of nonzero vectors is automatically linearly independent.

Gram-Schmidt Process

Given basis $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3\}$, produce orthonormal basis $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$:

$$\begin{aligned}\mathbf{v}_1 &= \mathbf{x}_1 \\ \mathbf{v}_2 &= \mathbf{x}_2 - \frac{\mathbf{x}_2 \cdot \mathbf{v}_1}{|\mathbf{v}_1|^2} \mathbf{v}_1 \\ \mathbf{v}_3 &= \mathbf{x}_3 - \frac{\mathbf{x}_3 \cdot \mathbf{v}_1}{|\mathbf{v}_1|^2} \mathbf{v}_1 - \frac{\mathbf{x}_3 \cdot \mathbf{v}_2}{|\mathbf{v}_2|^2} \mathbf{v}_2\end{aligned}$$

Then normalize: $\mathbf{u}_i = \mathbf{v}_i / |\mathbf{v}_i|$

QR Decomposition

$A = QR$ where Q has orthonormal columns (from Gram-Schmidt on columns of A) and R is upper triangular.

$$R = Q^T A \text{ (since } Q^T Q = I \text{)}$$

Orthogonal Projection

Projection of $\mathbf{y}$ onto subspace W with orthonormal basis $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$:

$$\hat{\mathbf{y}} = (\mathbf{y} \cdot \mathbf{u}_1) \mathbf{u}_1 + \dots + (\mathbf{y} \cdot \mathbf{u}_p) \mathbf{u}_p$$

Projection matrix: $P = QQ^T$ where $Q = [\mathbf{u}_1 \dots \mathbf{u}_p]$. Then $\hat{\mathbf{y}} = P\mathbf{y}$.

Least Squares

For overdetermined $A\mathbf{x} = \mathbf{b}$ (no solution), minimize $|\mathbf{b} - A\mathbf{x}|$:

$$\textbf{Normal equations: } A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$$

If A has linearly independent columns: $\hat{\mathbf{x}} = (A^T A)^{-1} A^T \mathbf{b}$

Example: Fit a line $y = \beta_0 + \beta_1 x$ to points $(0, 1), (1, 3), (2, 4), (3, 5)$.

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}, \mathbf{b} = \begin{pmatrix} 1 \\ 3 \\ 4 \\ 5 \end{pmatrix}$$

$$A^T A = \begin{pmatrix} 4 & 6 \\ 6 & 14 \end{pmatrix}, A^T \mathbf{b} = \begin{pmatrix} 13 \\ 26 \end{pmatrix}$$

Solve: $\hat{\beta}_0 = 1.5, \hat{\beta}_1 = 4/3 \approx 1.33$

6.7 Linear Transformations

Definition

$T : V \rightarrow W$ is linear iff:

1. $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$
2. $T(c\mathbf{u}) = cT(\mathbf{u})$

Equivalently: $T(c\mathbf{u} + d\mathbf{v}) = cT(\mathbf{u}) + dT(\mathbf{v})$ and $T(\mathbf{0}) = \mathbf{0}$.

Matrix Representation

Every linear $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ has $T(\mathbf{x}) = A\mathbf{x}$ where $A = [T(\mathbf{e}_1) \cdots T(\mathbf{e}_n)]$ ($m \times n$).

Standard transformations in $\mathbb{R}^2$:

- Rotation by θ : $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$
- Reflection over x -axis: $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
- Projection onto x -axis: $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$

Kernel and Image

$$\ker(T) = \{\mathbf{x} : T(\mathbf{x}) = \mathbf{0}\} = \text{null}(A)$$

$$\text{im}(T) = \{T(\mathbf{x}) : \mathbf{x} \in V\} = \text{col}(A)$$

- T is **one-to-one** (injective) iff $\ker(T) = \{\mathbf{0}\}$
- T is **onto** (surjective) iff $\text{im}(T) = W$

Change of Basis

If $B = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ and $B' = \{\mathbf{b}'_1, \dots, \mathbf{b}'_n\}$ are bases, the change-of-basis matrix from B to the standard basis is $P_B = [\mathbf{b}_1 \cdots \mathbf{b}_n]$.

$[T]_{B'} = P^{-1}[T]P$ where P is the change-of-basis matrix.

6.8 Matrix Decompositions

LU Decomposition

$A = LU$ where L is lower triangular (1's on diagonal) and U is upper triangular.

Obtained by tracking the row operations in Gaussian elimination. Used to solve $A\mathbf{x} = \mathbf{b}$ efficiently: forward solve $L\mathbf{y} = \mathbf{b}$, then back-substitute $U\mathbf{x} = \mathbf{y}$.

Singular Value Decomposition (SVD)

For any $m \times n$ matrix A :

$$A = U\Sigma V^T$$

- U : $m \times m$ orthogonal matrix (columns = left singular vectors — orthonormal basis for $\mathbb{R}^m$)

- Σ : $m \times n$ diagonal matrix with singular values $\sigma_1 \geq \sigma_2 \geq \dots \geq 0$ on diagonal
- V : $n \times n$ orthogonal matrix (columns = right singular vectors — orthonormal basis for $\mathbb{R}^n$)

Computing SVD:

- Singular values $\sigma_i = \sqrt{\lambda_i(A^T A)}$
- V = eigenvectors of $A^T A$
- U = eigenvectors of AA^T

Applications:

- Rank of A = number of nonzero singular values
- Pseudoinverse: $A^+ = V\Sigma^+U^T$ where Σ^+ replaces each nonzero σ_i with $1/\sigma_i$
- Least squares: $\hat{\mathbf{x}} = A^+\mathbf{b}$
- Data compression / low-rank approximation: retain top k singular values

Common Mistakes:

- Gram-Schmidt: not re-orthogonalizing against ALL previous vectors in each step
- SVD: mixing up left and right singular vectors (U vs. V)
- Least squares: applying normal equations when $A^T A$ is singular (use SVD instead)

End of Guide — Algebra · Calculus 1–3 · Differential Equations · Linear Algebra